# Yuxuan Gu

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## EDUCATION

**Imperial College London MEng Electrical and Information Engineering** **2021-2025**

- Dean's list for Year 2 (2022-2023)
- Dean's list for Year 1 (2021-2022) (top 5%)
- Modules: Circuits analysis, Analogue electronics, Digital electronics, Computer architecture, Control Systems, Signals and Communication, Programming, Mathematics
- Accomplished all the basic and challenging tasks in the lab and analysed tough questions with classmates in problem classes
- Formulated the schedules for the term-time project and organised meetings to discuss implementation details
- Demonstrated good time management skills, investigating literature involving control, robotics, electronics, and telecommunication in extra-curricular time

**Jinling High School, Nanjing, China** **2018-2021**

**A Levels**

- Accumulative marks Top 1 in China – November 2020 Cambridge Outstanding Learner Award (Best Across Three) and gave a speech as a Student Rep in the awards ceremony
- Further Mathematics (A*), Physics(A*), Chemistry(A*), Mathematics(A*)

## WORK EXPERIENCE

**Imperial College – Undergraduate Research Opportunities Programme (UROP)** **2023 summer**

- Developed a colour-tracking 4-DOF robotic arm using remote control within ROS2 framework
- Integrated a USB camera as a sensor within the feedback loop for effective bus servo motor management, while also employing geometric inverse kinematics analysis to calculate joint variables from end-effector position and orientation
- Employed a Raspberry Pi to transmit commands to the motors through UART, while concurrently publishing image frames to ROS2 topics
- Implemented controller separation onto a remote machine, enabling communication with the local node through the ROS2 network, and subsequently conducted an analysis of stability margins
- Built a 2$^{nd}$ order transfer function model by exploring step response of one motor

**Evotrack – Software engineer summer internship** **2022 summer**

- Analysed usage data of E-vehicles charging stations in Paris and ran several k-means clustering algorithms to divide stations into clusters
- Established a routine to obtain the number of given list of facilities around given addresses using Google Maps API
- Trained ensemble models (e.g., Gradient Boosting) to predict the future utilisation of a station based on the past usage data and number of surrounding facilities and achieved high accuracy
- Developed excellent programming skills and machine learning knowledge by exploring all challenging tasks independently

**Imperial College Careers Service Work Shadowing Scheme** **2022 summer**

- Experienced a day of life of an electrical engineer to maintain the water pumping systems
- Analysed pressure signal, PID feedback control of pump motor, and CHP units
- Compared parameters on the control panel in old-fashioned and modern plant rooms and discussed the development in this industry

## PROJECTS

**Imperial College London – Self-balancing autonomous maze-solving rover 2023**

- Developed a self-balancing rover that autonomously solves mazes, maps them in real-time, and identifies the shortest path
- Implemented a comprehensive system equipped with various components, such as NEMA 17 stepper motors, A4988 motor driver modules, an MPU-6050 gyroscope, an ESP32 microcontroller, an FPGA board, and a camera
- Leveraged FPGA vision technology to enhance image processing and analysis capabilities, enabling precise navigation and obstacle detection.

- Implemented sophisticated algorithms such as PID control, triangulation and A* algorithm, clustering methods, and graph theory to optimize rover performance and achieve efficient maze-solving strategies.
- Worked in a team of 2 EEE and 4 EIE people, assuming responsibility for FPGA vision, motor driving, PID control, and establishing seamless communication between esp32/PC and FPGA/esp32

**Imperial College London – Remotely-controlled rover 2022**

- Designed a rover that can identify rocks emitting unusual electromagnetic and acoustic signals
- Utilised Fourier Transform, filtering methods and modulation & demodulation techniques
- Sharpened practical and programming skills by debugging faults in circuits and Arduino codes
- Collaborated with 5 teammates and responsible for all signal identifications

**Light-seeking robot 2021**

- Successfully created a robot prototype that moves towards the light
- Carried out soldering PCB, programming to Arduino (C++), and testing the robot in a team of 2
- Employed electronic technology and motion control technique
- Demonstrated the functions to academics and tutor group members

**AI-based plants water shortage warning device 2020**

- Developed research methodologies by exploiting online resources
- Developed a model to match the image of leaves with soil humidity
- Came up with the idea of using AI to remind ordinary people to water the plant
- Collected image of leaves and soil humidity data using Raspberry Pi and Arduino

**Haihua AI Challenge -- Garbage Classification 2020**

- Achieved 4th place in the leader board
- Trained a CNN model to classify 80,000 rubbish images into 198 corresponding categories
- Distributed sub-tasks to 2 of my teammates and launched into model building and training using Python with the help of senior students and online resources

## TECHNICAL SKILLS

- Programming language: C++, Python
- Simulation tools: MATLAB, LTSpice
- Microcontroller and embedded system: Arduino, Raspberry Pi
- Language: English (IELTS: overall 7.5 with each band no less than 7.0), Chinese (native)

## EXTRA-CURRICULAR INTERESTS

**Badminton**

- Active member of Imperial Badminton Club, attending social sessions every week
- Taught badminton beginners with patience

**Violin**

- Has been to many violins solo and orchestra concerts
- Played a violin solo at concerts
- Collaborated with friends from time to time.
  E.g., https://www.youtube.com/watch?v=jy4wHAXgEWI

## VOLUNTEERING

**Video shooting** **2021 -- now**

- Voluntarily shot some videos introducing college lives to junior students in my high school
- Played some violin pieces to share the beauty of music to people online
- For more information, please visit https://space.bilibili.com/594818982

**Charity** **2020**

- Donated 500RMB to the Project Hope to support the anti-virus battle
- For more information, please visit https://guyuxuan9.github.io/2020/01/31/My-participation-in-2019-nCoV-Prevention-Treatment-Project/